

Drone Payload Structure Design and Analysis

In this project, you are optimizing drone arm design, including geometry and material selection, to maximize payload capacity while ensuring structural integrity under motor-induced loads. You will use MATLAB to perform force balance analysis and finite element modeling to quantitatively support your design solution.

Use this Live Script as a template to get started on your project solution. You may include helper functions at the bottom of this document or as separate files. For a tutorial on how to use Live Scripts, check out this [video](#).

Suggested Tasks

1. Gather your starter assumptions and load the material properties into MATLAB.
2. Propose at least two drone arm designs
3. Perform a thrust-to-weight analysis to evaluate each of your designs across all material options. Your design must meet a minimum payload requirement and safety standards.
4. Perform finite element analysis across all design and material options.
5. Recommend a final design solution.

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Break Down the Problem

In the space below, define the problem statement:

The objective of this project is to test different drone arm designs and material options for a quadcopter drone. The goal is to maximize payload capacity while maintaining structural integrity under motor induced loads. MATLAB will be used to perform thrust-to-weight analysis and finite element analysis to compare each design and material combination, allowing us to choose the best design that meets the design requirements.

List the requirements, constraints, and success criteria for the project:

Requirements

- Design and model two or more drone arms for a quadcopter drone
- With given material options, perform thrust to weight analysis across each option to see which material optimizes payload capacity while meeting safety requirements
- Perform finite element analysis on each drone arm's 3D model in the form of an .stl file in MATLAB from a combination of upward motor thrust force and downward motor weight force to obtain its maximum von Mises stress, displacement, and safety factor.
- Compare and contrast material options to see which arm meets the design requirements based on the Von mises stresses, displacements, and safety factor from the FEA

Constraints

- Safety factor must be at least 1.5
- Assuming 1kg of thrust per motor, drone must be able to support a minimum 0.5kg payload
- Thrust to weight ratio must be at least 2:1

Success Criteria

- At least two drone design sketches showing key geometric features
- Results of thrust-to-weight analysis presented either in a summary table or plot
- Results of finite element analysis
- Numerical values presented in a summary table
- Visualizations of x-, y-, and z-displacement and Von Mises stress
- A final design recommendation justified using the results of your analyses

What is your proposed solution?

We propose to make a MATLAB live script that can automatically analyze multiple drone arm designs in the form of .stl files, allowing the user to know each design's maximum payload, average material cost, and structural integrity under motor-induced loads for each user-defined material combination.

List the steps you will need to take to achieve your proposed solution:

1. Gather starter assumptions and define material properties into MATLAB
2. Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features
3. Perform a thrust-to-weight analysis across all design and material options to evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements
4. Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. (To simplify the task, we will perform the FEA only on a single drone arm.)
5. Using the results of both the thrust-to-weight analysis and finite element analysis, propose a final design solution for the drone arm that maximizes payload capacity while meeting safe flight standards and maintaining structural integrity under load

What challenges do you face in the process of achieving the solution?

- Creating a real-life scale drone arm in CAD
- Creating a user-interactable live script in MATLAB that meets all of the design requirements
- Presenting data in a user-friendly format in MATLAB
- Troubleshooting code issues
- Coordinating with teammates and effectively using GitHub

Task 1: Gather your starter assumptions and load material properties

Load the provided `droneArmMaterials.mat` file into MATLAB. This contains a data struct with the material properties for six different materials commonly used for drones (carbon fiber composite, aluminum alloy, fiberglass composite, PLA plastic, ABS plastic, and wood). For each material, you have been provided the following information:

- Density (ρ), in $\frac{\text{kg}}{\text{m}^3}$
- Young's Modulus (E) in Pa
- Poisson's ratio (ν), unitless
- Yield strength, in Pa
- Average cost per meter in US dollars

For this project, you can assume the following design parameters:

- Quadcopter drone with 4 motors and 4 propellers
- Each motor can provide up to 1kg thrust
- Acceleration due to gravity: $g = 9.81 \frac{\text{m}}{\text{s}^2}$
- Total weight of the drone components without the arms or the payload: 1kg

The breakdown of the weight of each drone component is as follow:

- Battery = 0.3kg
- Each motor = 65g, 4 motors = 0.26kg
- Each propeller = 10g, 4 propellers = 0.04 kg
- Each electronic speed controller = 30g, 4 electronic speed controllers = 0.12 kg
- Flight controller and electronics = 0.05kg
- Receiver/GPS = 0.03kg
- Drone frame (center body) excluding arms = 0.2kg

Design Parameters

Before you run the code, make sure your current directory is set to the project folder!

Drone Component Masses

Key Values

Total maximum thrust = 39.20 N

Base drone mass check = 1.00 kg

Minimum payload requirement = 0.50 kg

Required thrust-to-weight ratio = 2.0:1

Task 2: Propose at least two drone arm designs

Develop at least two distinct drone arm designs by drawing clear, neatly labeled sketches that show key geometric features, including:

- Overall shape (e.g. beam, tube, truss structure)
- Dimensions (length, width, thickness)
- Motor mount location
- Where the arm connects to the drone body

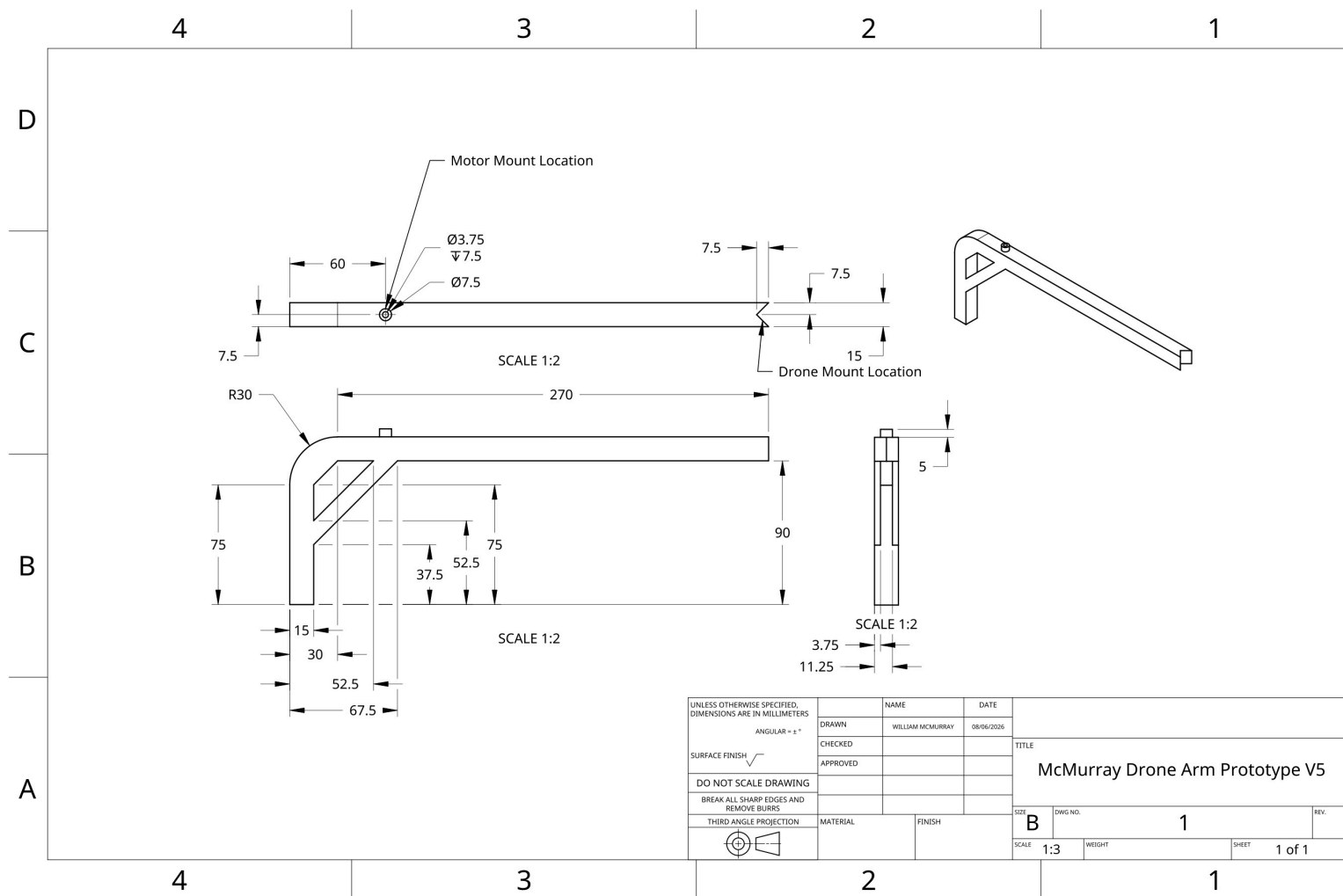
The two designs should be meaningfully different from each other (e.g. different geometries or structural strategies) and not just small variations of the same idea.

Navigate to the 'Insert' tab and insert pictures of your sketches in the space below (hand-drawn or digital).

Include a short design description for each sketch that includes:

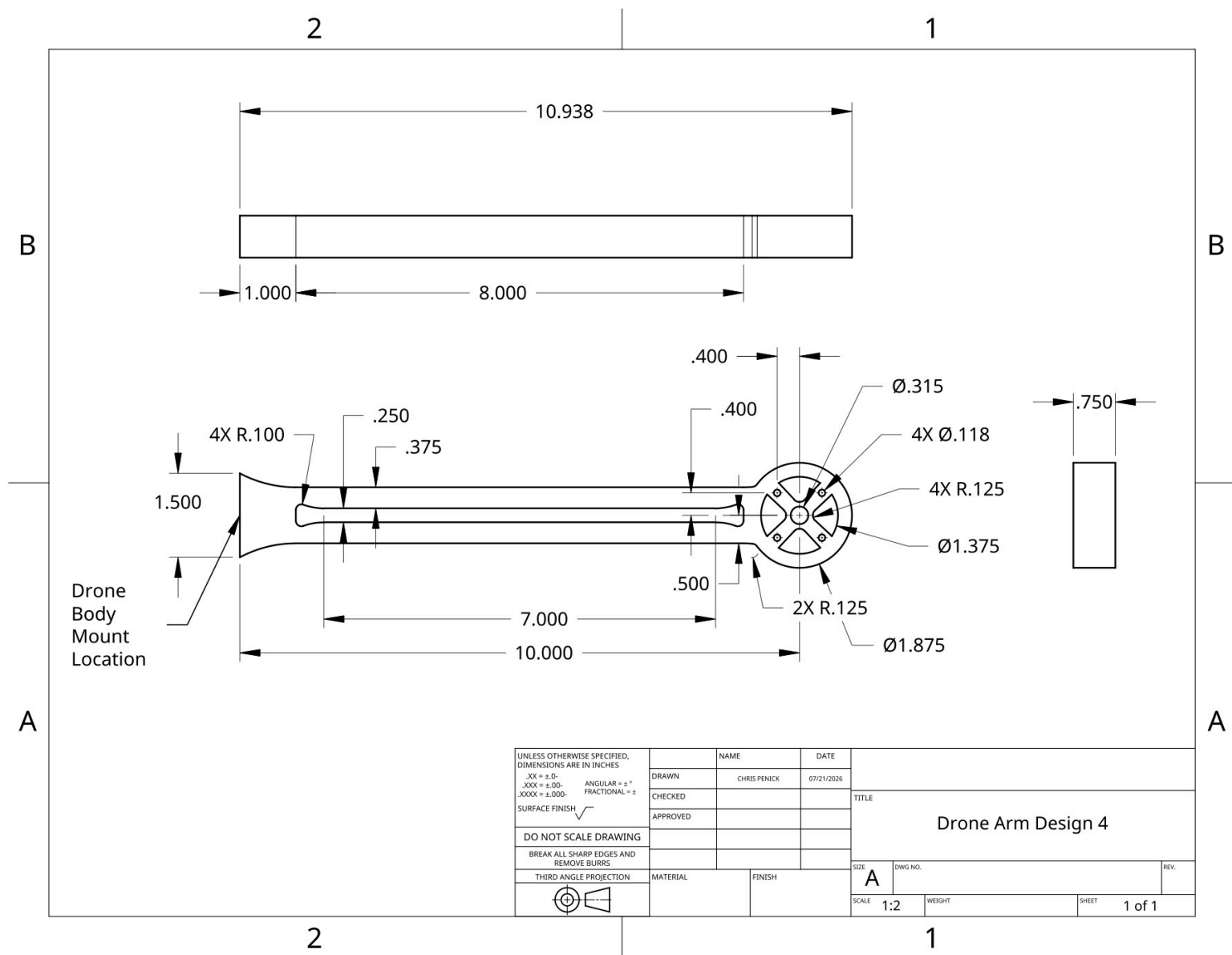
- Geometry (e.g. solid beam, hollow tube, beam with trusses)
- Why you chose this design
- How it might perform under load

McMurray_DroneArmPrototypeV5



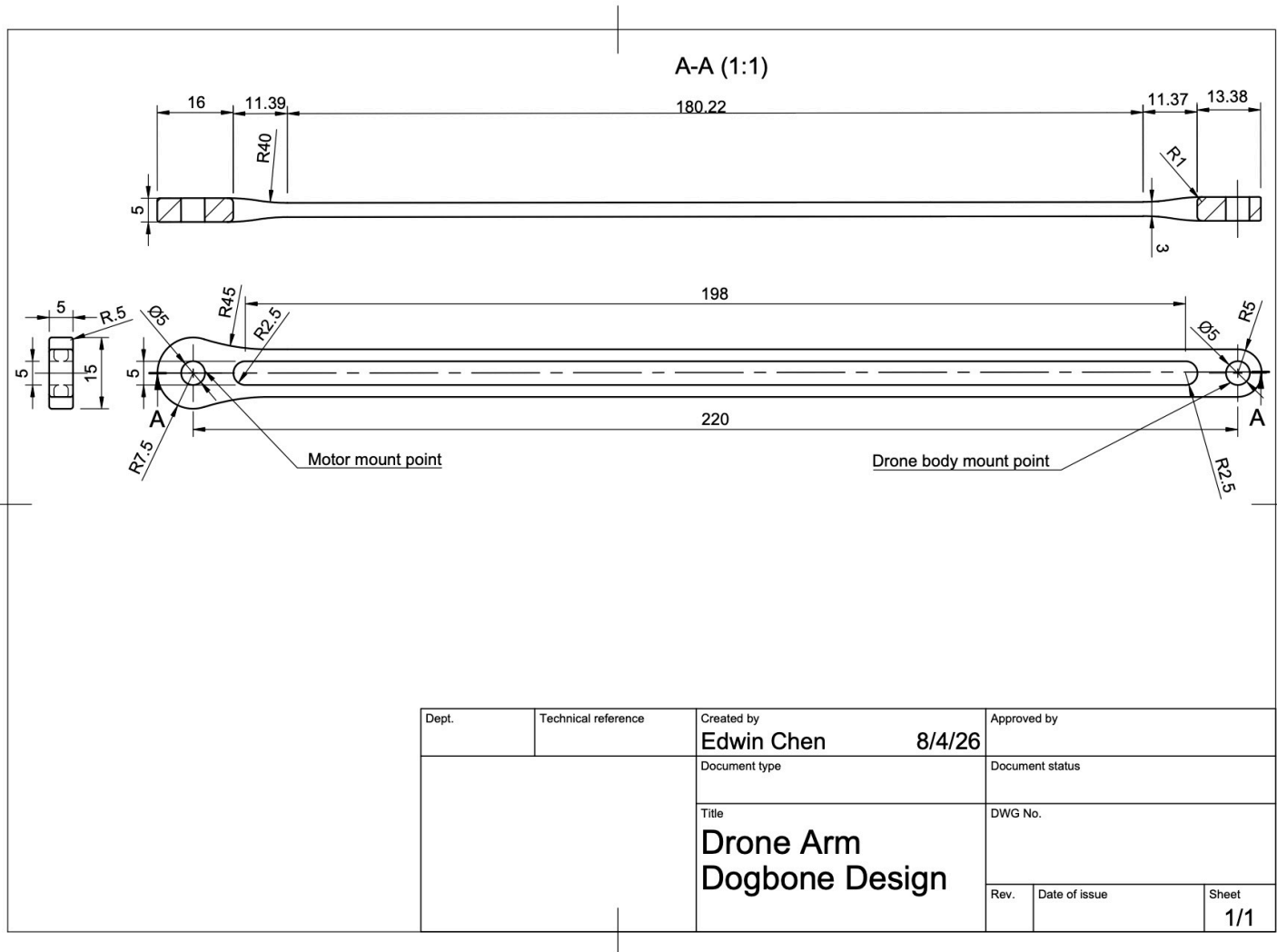
My design is a long drone arm that bends down into the leg with a fillet on the outside and a chamfer on the inside in order to remove and distribute any additional stress that might result from sharp corners. An additional pair of ribs connect the arm and leg for additional stability to support the arm that will attach to the drone holding the payload. The motor mount extends out from the arm cylindrically and has a smaller 3.75mm cylindrical hole for the motor to be placed in. The drone arms will mount to each of the four corners of a square shaped drone body and will extrude diagonal to each face of the body. Under the load of the motor, the diagonal beam connecting the arm and the leg will take away some of the stress from going to the long arm. However, PLA plastic will provide the least effective arm in terms of displacement because of its high elasticity and low strength. -Will

chris_final_drone_arm



I chose a design that was a solid beam with a cutout slot in the middle to reduce the weight and increase the airflow through the arm. I chose to do a solid design to minimize complexity while also ensuring that it would be strong enough to support the payload and the thrust from the motor. I designed the length of the drone arm to be 280 mm in order to be long enough to support propellers of 127 mm (5 inches), which would be long enough to lift the body of the drone and payload. When loaded the beam will perform well and not undergo much displacement with most of the materials. PLA will have the most displacement due to the softness of the plastic. -Chris

DroneArm_Dogbone



This drone arm design is a narrow and solid beam with dimensions similar to what's expected for a fast-flying FPV drone. I chose this design because the cut and slimming taper in the middle will save weight and material, while the motor and drone body mount points' has extra material to increase the amount of threading available and act as additional reinforcement to improve long-term durability (The dogbone name comes from the middle taper in the drone arm's side profile). Sharp corners were eliminated through fillet operations to decrease stress concentrations, giving it a round design. Even so, this arm design will suffer from bending and torsional stresses under a motor's load because of the lack of material and bracing in the arm's tapered portion. Therefore, this design warrants the usage of a very stiff material like a carbon-fiber reinforced polymer composite to minimize the bending and torsional stresses and maximize its effectiveness. -Edwin

Task 3: Perform thrust-to-weight analysis across all design and material options

Write a MATLAB function that will perform a thrust-to-weight analysis across all design and material options. This analysis will help you evaluate which drone arm design and which material optimizes payload capacity while meeting suggested safety requirements.

- Your design must support a minimum payload of 0.5kg.
- Your design must meet a thrust-to-weight ratio of at least 2:1 to ensure safe and stable flight. This safety standard is widely used in drone design. Conceptually, it means that the drone's maximum thrust must be at least twice its weight. This ensures stable hovering and maneuvering with a safety margin. Note that the thrust-to-weight safety margin is a system-level safety standard for drone flight that is distinct from the structural safety factor of the drone arms (which you will evaluate later through finite element analysis).
- Your goal is calculate maximum payload capacity for each design and material option (i.e. what is the heaviest payload your drone design could carry while still allowing the drone to safely take off?)

Ensure that the results of your analysis can be readily interpreted, e.g. in the form of a summary table or plot.

Note: if you have two drone arm designs, the expected output should be 12 payload capacity estimates (2 designs across 6 materials options).

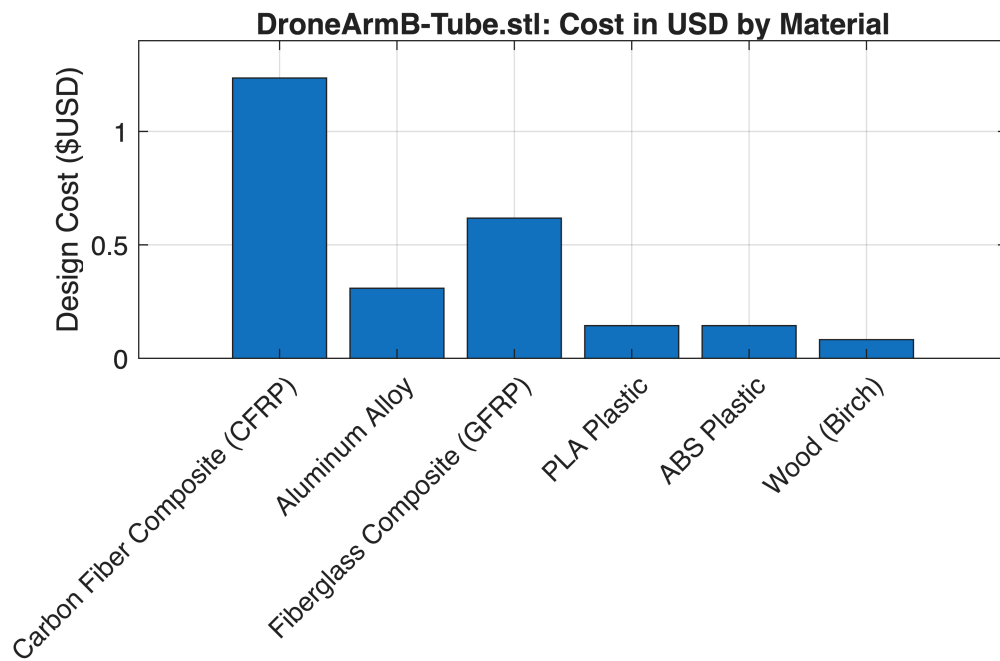
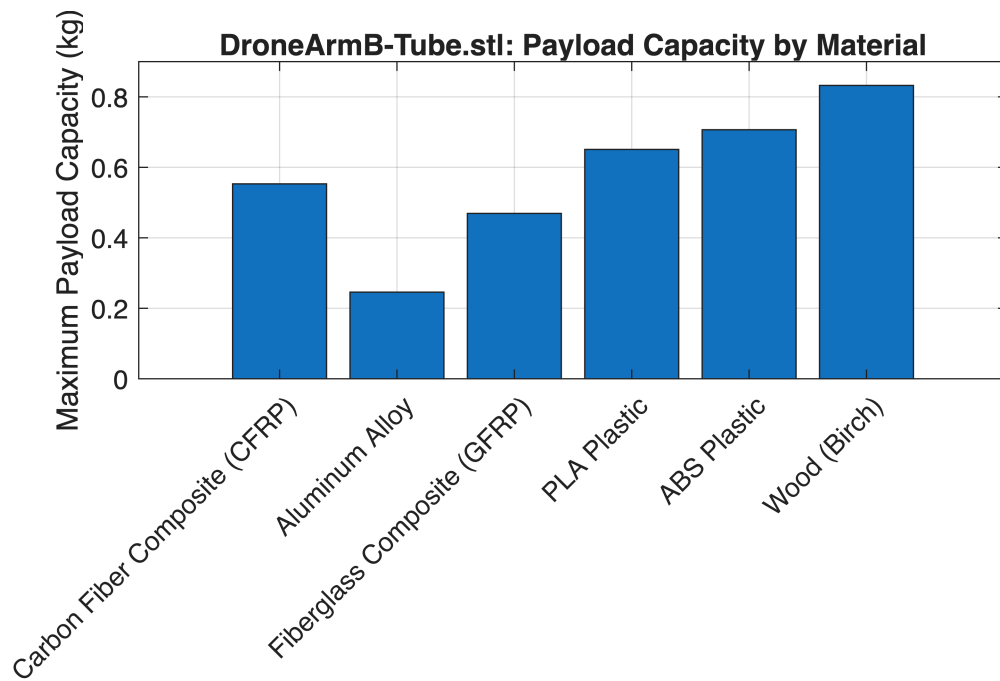
Maximum allowed total drone mass = 2.00 kg

Make sure all the files you want to analyze is located in the resource/stl folder!

Thrust-to-Weight Analysis Results

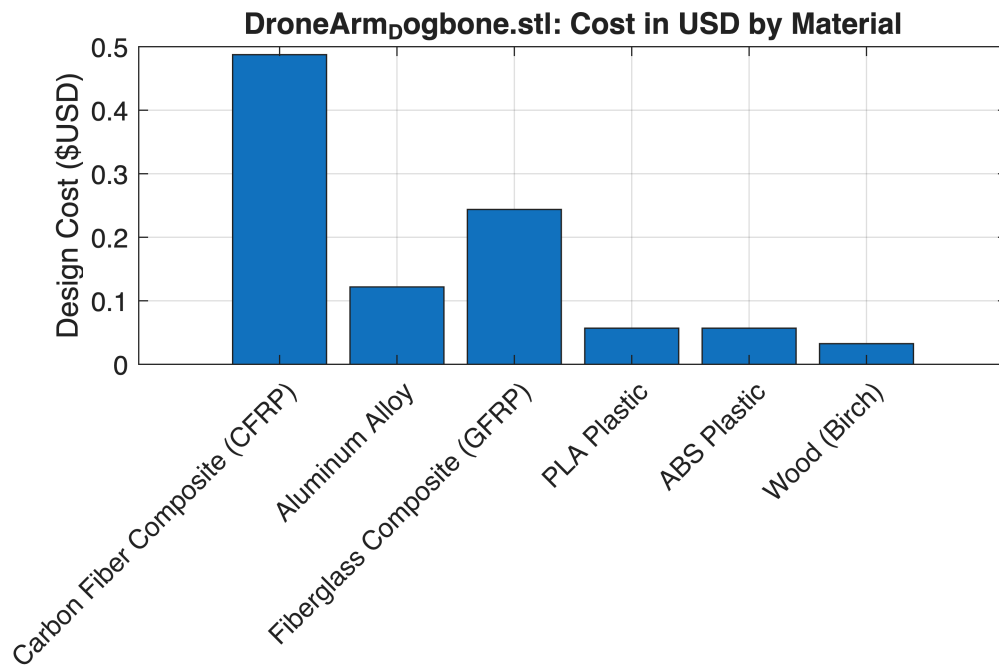
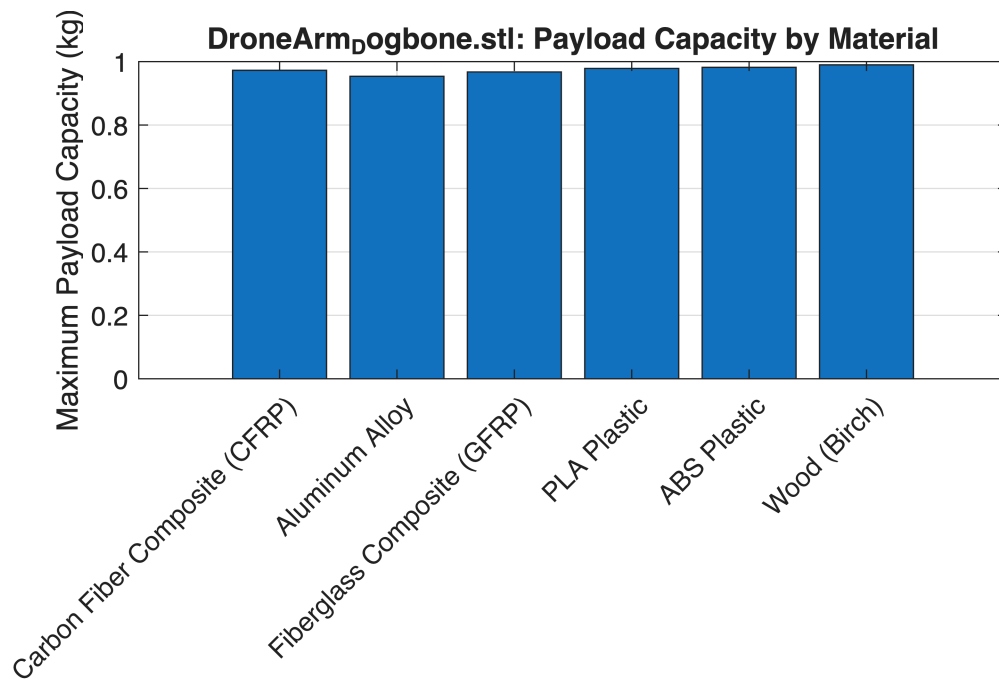
Analyzing DroneArmB-Tube.stl...
ans = 6x5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.1117	0.4470	0.5530
2	"Aluminum Alloy"	0.1886	0.7543	0.2457
3	"Fiberglass Composite (GFRP)"	0.1327	0.5308	0.4692
4	"PLA Plastic"	0.0873	0.3492	0.6508
5	"ABS Plastic"	0.0733	0.2933	0.7067
6	"Wood (Birch)"	0.0419	0.1676	0.8324



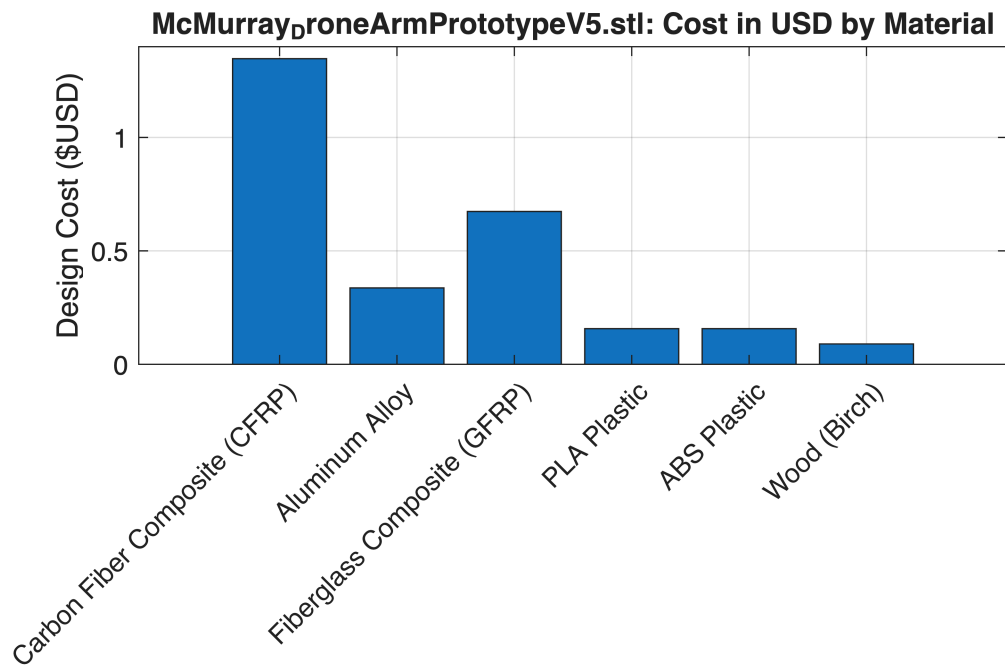
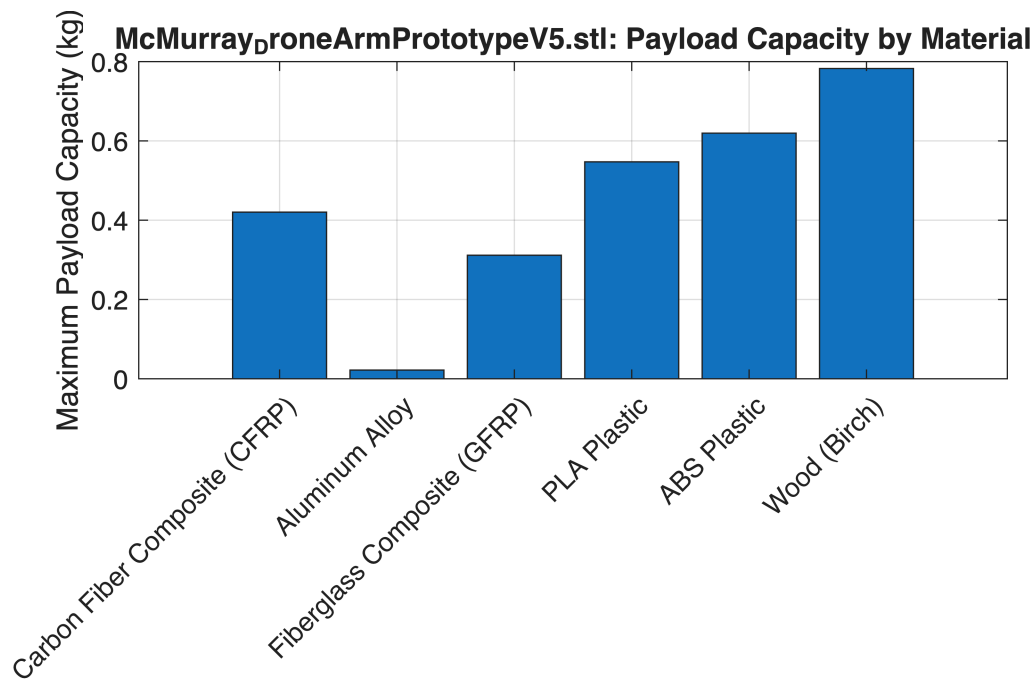
Analyzing DroneArm_Dogbone.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.0069	0.0274	0.9726
2	"Aluminum Alloy"	0.0116	0.0463	0.9537
3	"Fiberglass Composite (GFRP)"	0.0081	0.0326	0.9674
4	"PLA Plastic"	0.0054	0.0214	0.9786
5	"ABS Plastic"	0.0045	0.0180	0.9820
6	"Wood (Birch)"	0.0026	0.0103	0.9897



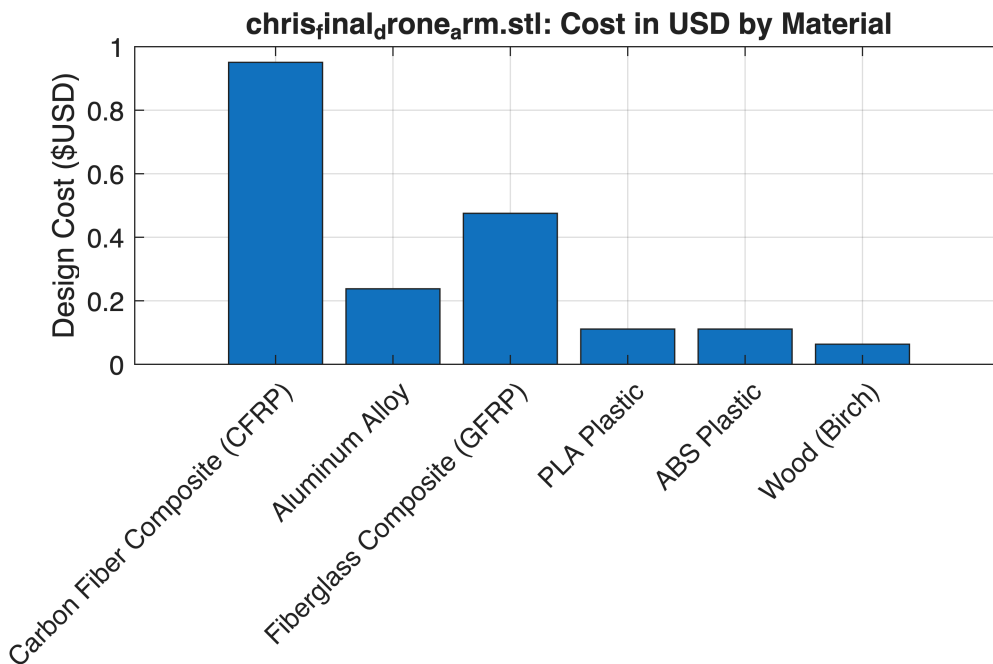
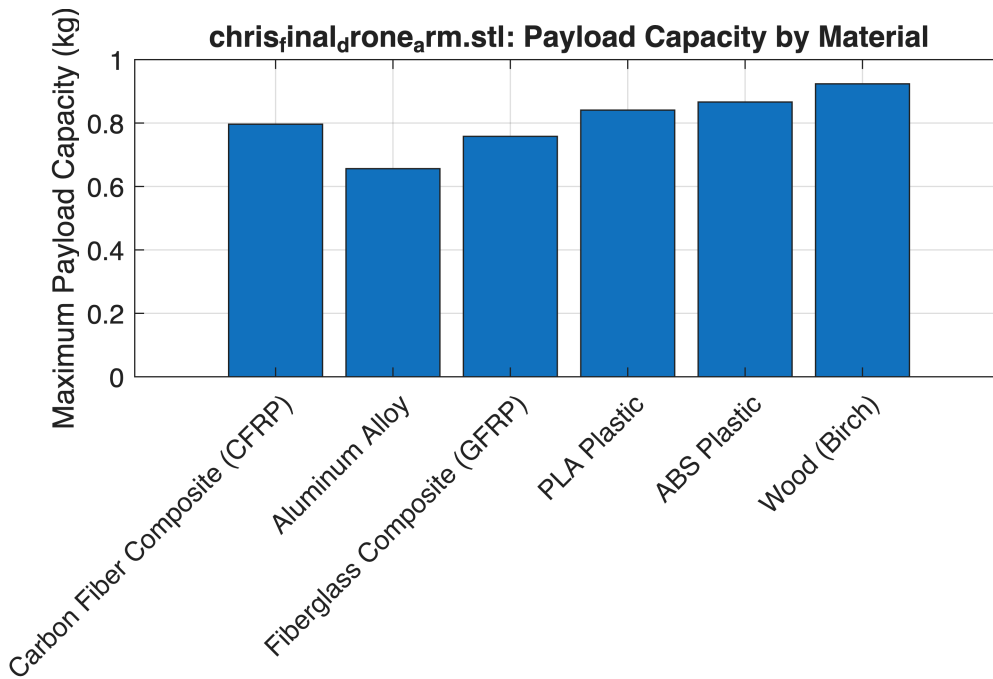
Analyzing McMurray_DroneArmPrototypeV5.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.1449	0.5796	0.4204
2	"Aluminum Alloy"	0.2445	0.9781	0.0219
3	"Fiberglass Composite (GFRP)"	0.1721	0.6883	0.3117
4	"PLA Plastic"	0.1132	0.4528	0.5472
5	"ABS Plastic"	0.0951	0.3804	0.6196
6	"Wood (Birch)"	0.0543	0.2174	0.7826



Analyzing chris_final_drone_arm.stl...
ans = 6×5 table

	Material	OneArmMass_kg	TotalArmMass_kg	payloadCapacity_kg
1	"Carbon Fiber Composite (CFRP)"	0.0509	0.2037	0.7963
2	"Aluminum Alloy"	0.0860	0.3438	0.6562
3	"Fiberglass Composite (GFRP)"	0.0605	0.2419	0.7581
4	"PLA Plastic"	0.0398	0.1592	0.8408
5	"ABS Plastic"	0.0334	0.1337	0.8663
6	"Wood (Birch)"	0.0191	0.0764	0.9236



Task 4: Perform finite element analysis across all design and material options

Perform finite element analysis (FEA) across all design and material options to evaluate the structural safety factor of your drone arm design. To simplify the task, you will perform the FEA only on a single drone arm.

Using a CAD software of your choice, first create a 3D model of one drone arm for each of your designs. Be sure your model can be exported as a STEP or STL file for analysis in MATLAB.

- Recommended CAD software options: Tinkercad, Onshape, Fusion 360, Solidworks (most have free versions)

Import your model into MATLAB and perform FEA. In your FEA model, you should model and apply two loads to the drone arm: upward thrust force from the motor and downward weight of the motor.

- Helpful resources: [online tutorial](#), documentation for [Structural Mechanics](#) and [Partial Differentiation Equation Toolbox](#), and [example workflow](#)

For each design and material combination, you should report the overall maximum displacement, maximum stress, and factor of safety and include visualizations of the x-, y-, and z-displacement and Von Mises stress. Ensure that your results are presented for easy interpretation (e.g. present a table with numerical results, clearly labeled subplots for the visualizations).

- The factor of safety compares how strong a material is to how much stress it experiences, indicating whether a design is safe under the given loads. For this project use the equation:

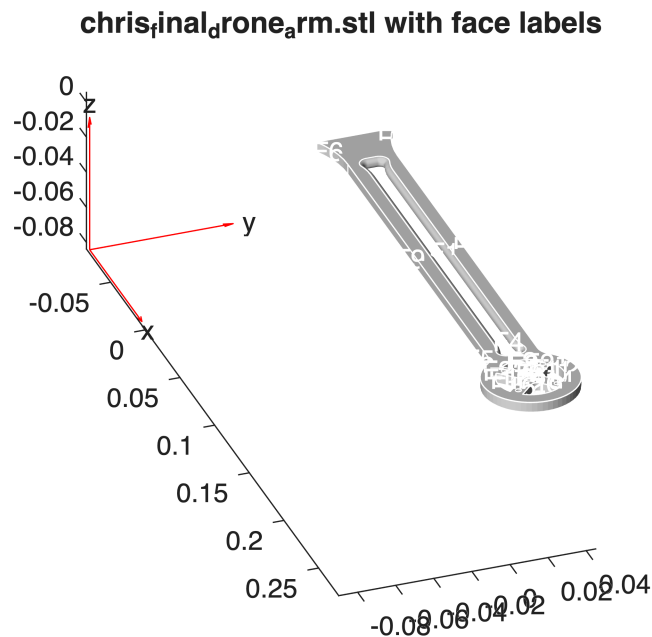
$$\text{FoS} = \frac{\text{Material Yield Strength (Pa)}}{\text{Maximum von Mises Stress}}$$

Yield strength indicates how much stress a given material can withstand before permanently deforming. Maximum von Mises stress (derived from your FEA model) is a measure of how much stress your design experiences under the applied loads.

- A typical engineering safety margin for FoS is ~1.5-2+
- This [example](#) demonstrates the plots you are asked to include in your results

Choose the file to perform FEA on

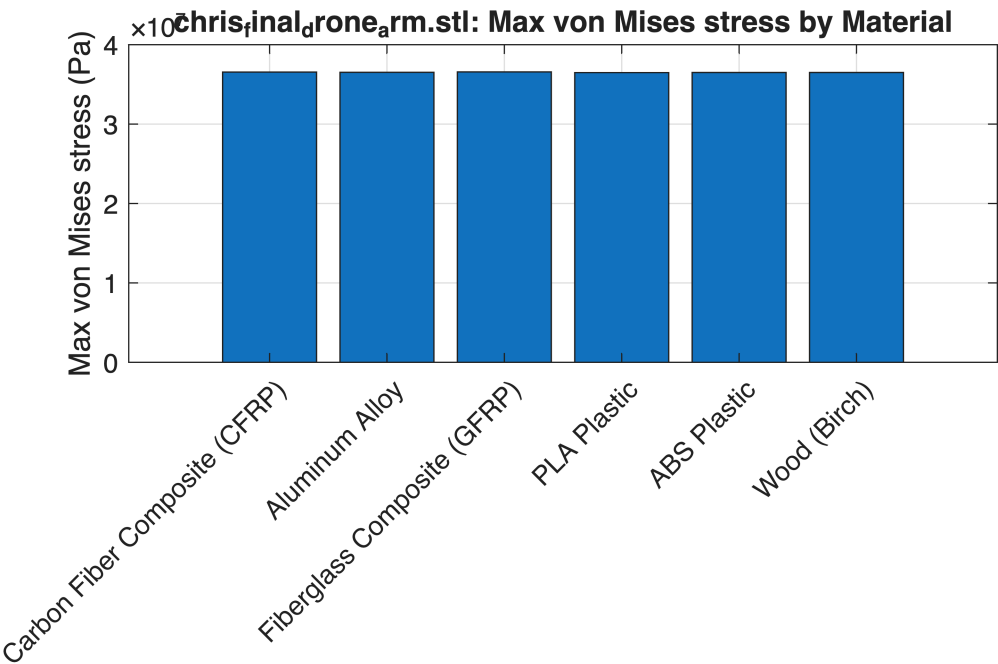
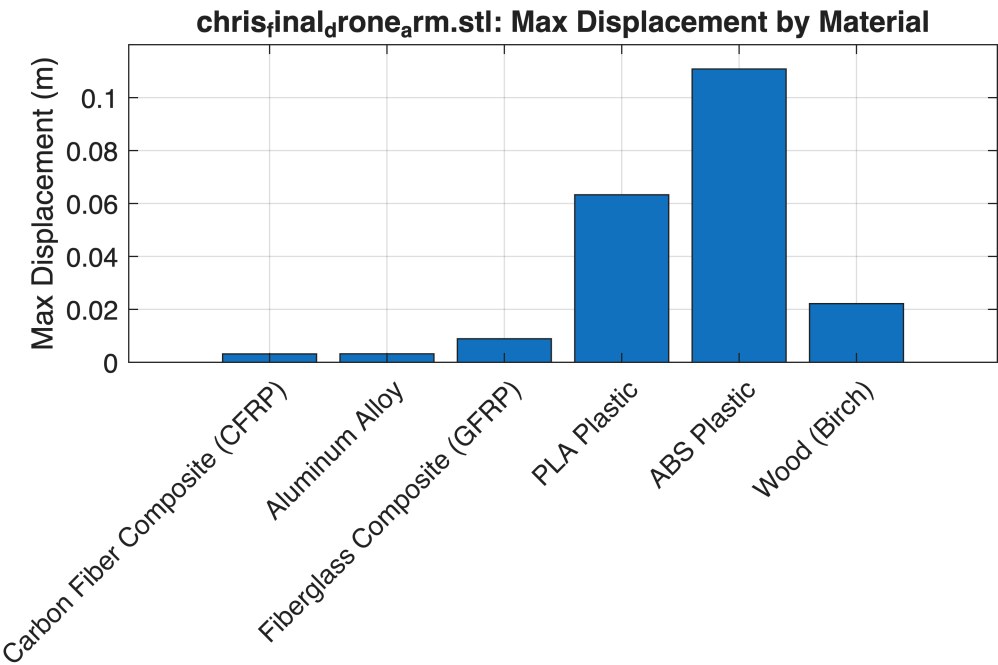
Modify view angle of visualizer

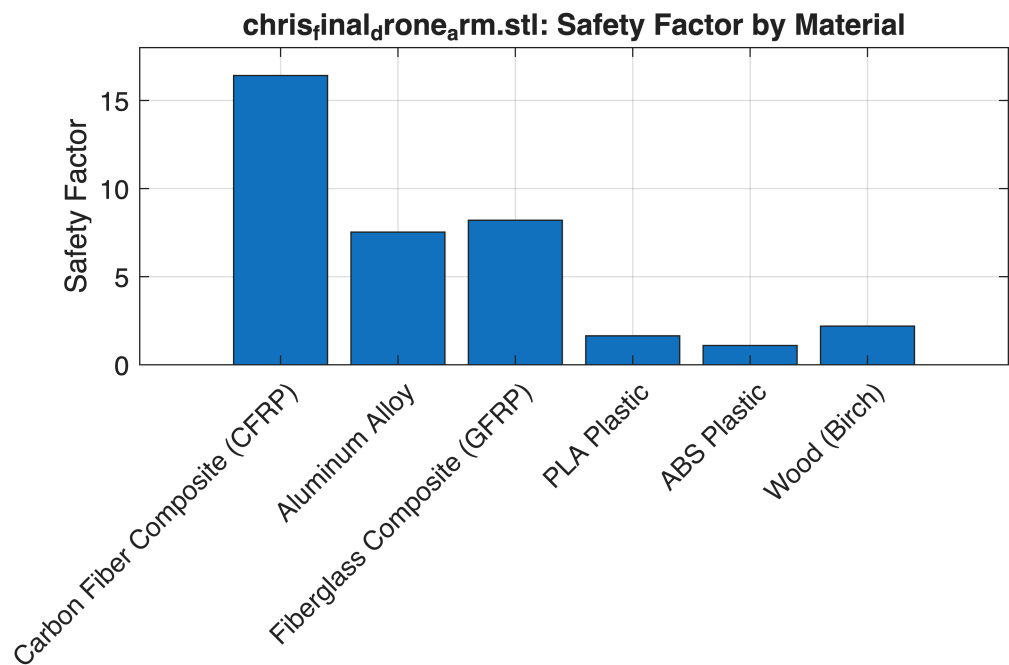


View the visualized model and identify the static and loaded face

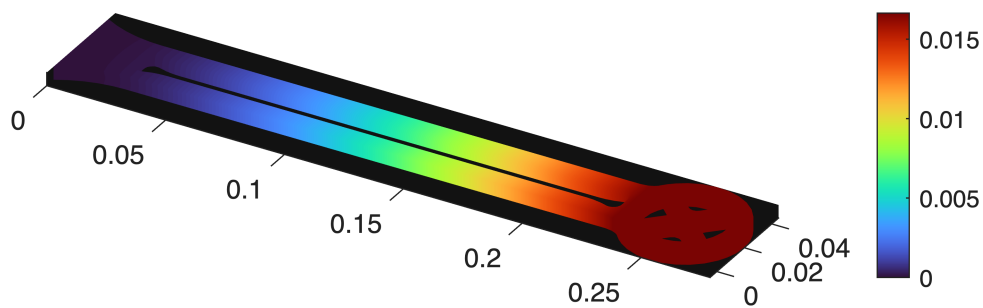
Warning: Some input points are not referenced by the triangulation.

FEA Results





Select material to visualize the PDE results



Final Summary Table

resultsFEA_Table = 6×6 table

...

	Material	maxDisplacement_m	maxVonMisesStress_Pa
1	"Carbon Fiber Composite (CFRP)"	0.0032	3.6552e+07
2	"Aluminum Alloy"	0.0032	3.6519e+07
3	"Fiberglass Composite (GFRP)"	0.0089	3.6574e+07

	Material	maxDisplacement_m	maxVonMisesStress_Pa
4	"PLA Plastic"	0.0633	3.6486e+07
5	"ABS Plastic"	0.1108	3.6497e+07
6	"Wood (Birch)"	0.0222	3.6497e+07

Elapsed time is 43.260253 seconds.

Interpretation of Results - Final Recommended Design Solution

Using the results of your thrust-to-weight analysis and finite element analysis, recommend a final drone arm design that meets all required constraints. Your recommendation should clearly identify the selected design (arm geometry and material choice), make use of quantitative results to justify your decisions, demonstrate that the design meets the constraints, explain the trade-offs considered, and include comparisons to alternative designs.

Based on our data, we recommend the usage of chris_final_drone_arm made with birch wood to be used as a drone arm. The design exceeds the minimum 0.5kg payload capacity with a payload capacity of 0.92kg; exceeds the recommended safety factor range of 1.5-2 by having a safety factor of 2.2; and has the lowest material cost of \$0.06 out of our material options, with the average material cost of \$0.3232 and the most expensive material option being \$0.950.95 for a carbon fiber polymer composite. However, the arm has a maximum displacement of 22mm, considerably higher than aluminum alloy and carbon fiber polymer composite's maximum displacement of 3mm. Despite the high displacement, the birch wood drone arm still exceeds the recommended safety factor range, allowing us to still consider the usage of that material for a drone arm.

In contrast, DroneArm_Dogbone made with carbon fiber reinforced polymer was our second-best option: It has a 0.97kg payload capacity, a safety factor of 1.7, costs \$0.49 in material, and has a maximum displacement of 47mm. Despite the slightly higher payload capacity, the dogbone design comes with too many drawbacks compared to chris_drone_arm_v3. The safety factor, maximum displacement, and material cost was considerably worse than chris_final_drone_arm made of birch wood, despite the use of a premium engineering material, so we ruled out DroneArm_Dogbone.

Thus, chris_final_drone_arm made of birch wood is the most feasible drone arm design that we've analyzed. The design's high cost efficiency makes it very suitable to mass produce for low-cost drone units.

What are some limitations of your work?

Generally, simulations can only predict reality through approximations, but cannot accurately reproduce it. In a similar manner, the FEA results will deviate from the drone arm's real-world behavior due to the assumptions and scope we worked with.

For example, the FEA doesn't take the material's manufacturing process into consideration. If the drone arm was made through FDM 3D printing, then its structural characteristics will be anisotropic, while producing the drone arm through subtractive manufacturing techniques like CNC or injection molding would give it isotropic

structural characteristics. While it's true that there could also be defects from its manufacturing process that can lower the arm's actual yield strength from the theoretical, taking the drone arm's safety factor into account during our analysis compensates for that. Altogether, our FEA most accurately reflects the structural characteristics of a drone arm not made from FDM 3D printing because our solution doesn't take anisotropy into account.

Additionally, our FEA solution doesn't account for the vibration and variable loading from real-world operating conditions. Throughout its service life, the drone would likely go through hard acceleration from takeoff and corrective maneuvers, subjecting the drone arm to cyclic stresses. The cyclic stresses will cause fatigue and increase the likelihood of failure over time. Meanwhile, propellers can cause resonance from motor harmonics within the drone body during usage. If a propeller vibrates a drone arm close to its natural frequency, it can compound its likelihood for failure, especially if the drone has gone through a substantial number of loading cycles. Because of that, our work is best for predicting a drone arm's short-term performance from a static load and further work will be needed to predict the longer-term durability of our drone arm.

Additionally, the design cost analysis doesn't take the manufacturing process into consideration, but rather the cost of the raw material for the drone arm volume. While it's true that the manufacturing expenses are indirectly reflected in the relative cost differences of each material selection, each of our proposed designs have unique geometry and pose different types of manufacturing challenges that indirectly increase the overall cost of the design. In other words, two different drone arm designs made with the same material with the same volume as each other will likely have different costs because, for example, one design might be more complex to machine than the other.

What are practical next steps?

After the FEA step, we will go back to the CAD phase and model real-world threads and ensure enough clearance for our electronic components. We will also verify that our design would be manufacturable with the manufacturing technique that we've loosely considered while creating our first design drafts. For example, since our drone arm will be made of birch wood, we'll have to verify if we're able to recreate our geometry visualized in the CAD phase given the tooling available for making a wood component.

Next, we'll build real-world prototypes of our drone. Our first prototype will be quickly 3D printed with PLA or ABS plastic or laser cut to verify fitment and clearance of all of our components, but it won't actually be functional. Afterwards, we will build the drone arm out of our chosen material to conduct laboratory tests to verify the FEA results. We will also test for motor harmonics, which our FEA solution doesn't cover.

If real-world tests verify that our drone arm design meets our design specifications with the entire drone, we will create the bill of materials and engineering drawings with GD&T so the drone can be mass-produced by suppliers.